

Phishing Email Detection & Awareness Report

A simulated training analysis of safe, suspicious, and phishing email patterns

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ASSIGNMENT Task 2 - Phishing Email Detection & Awareness System	DELIVERABLE Professional awareness report with simulated examples

Assessment Boundary

All content in this report is simulated for educational and portfolio purposes. The report uses fictional .example domains, non-functional links, and no real accounts, credentials, or inbox data.

CLASSIFY Recognise the differences between Safe, Suspicious, and Phishing messages.	ANALYSE Review sender domains, urgency, links, greetings, and sensitive requests.	RESPOND Verify independently, report suspicious messages, and protect accounts.
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1. Executive Summary

This report documents a phishing email detection and awareness exercise using simulated messages. The analysis compares a safe email, a suspicious account-verification email, and a phishing payment-warning email to demonstrate how users can recognise common deception techniques before interacting with a message.

The review focuses on visible and practical indicators: sender domain patterns, greeting style, urgency, link behaviour, requests for sensitive information, attachments, and overall message quality. The purpose is not to test or interact with real phishing campaigns; it is to strengthen awareness and safe decision-making.

Objectives

- Classify emails as Safe, Suspicious, or Phishing using observable indicators.
- Explain the potential impact of phishing in simple, business-friendly language.
- Provide prevention guidance that users can apply before replying, clicking, downloading, or sharing information.
- Create a documented portfolio deliverable supported by simulated training examples.

Scope and Ethical Boundaries

IN SCOPE Simulated emails, sender-domain review, visible links, message tone, awareness guidance	OUT OF SCOPE Real inbox access, malicious URL access, attachment execution, credential entry, or malware analysis
EVIDENCE BASIS Two simulated training visuals created for this internship task	CLASSIFICATION BASIS Sender, tone, links, greeting, sensitive requests, and overall authenticity

Important Note

A message should not be considered safe solely because it looks professional. When a request is unexpected, independently verify it through the organisation's official website, app, or support channel.

2. Methodology and Classification Model

Each simulated message was reviewed using a simple, repeatable process designed for everyday users. The emphasis is on pausing before action and looking for multiple indicators rather than trusting one detail alone.

1	2	3	4
Review sender	Read the message	Inspect links	Decide safely
Inspect the full address and domain.	Check greeting, tone, pressure, and context.	Hover over links and compare destination domains.	Verify independently, report, or delete.

Email Classification Framework

Classification	Observed Pattern	Recommended User Action	Risk Level
Safe	Clear sender domain, expected context, no pressure or sensitive request.	Normal caution. Verify if the message was unexpected.	Low
Suspicious	Lookalike domain, generic greeting, unusual urgency, or unclear link.	Do not click. Verify through official channels.	Medium
Phishing	Fake/unrelated domain, threatening language, sensitive request, deceptive link.	Do not reply, click, or download. Report and delete.	High

Indicators Reviewed

- Sender address and domain similarity
- Personalised versus generic greeting
- Tone, urgency, and threats
- Link destination and call-to-action behaviour
- Unexpected attachments and requests for passwords, OTPs, or payment details
- Overall quality, context, and consistency of branding

3. Simulated Email Comparison

Figure 1 presents the three simulated examples used in this report. Each example has a different risk profile and is intentionally labelled to support awareness training. The examples use fictional .example domains and non-functional links.


SIMULATED TRAINING EMAILS – No real accounts, links, or credentials used


These are examples created for educational purposes to demonstrate how to identify safe, suspicious, and phishing emails.

SAFE vs SUSPICIOUS vs PHISHING

1. SAFE EMAIL

From: orders@northstar.example ✓

To: keerit.kapoor@example.com

Subject: Your NorthStar Order Confirmation

Date: 20 June 2026, 10:30 AM

★ NORTHSTAR

Hi Keerit Kapoor,

Thank you for your order with NorthStar!

Your order #NS-58921 has been confirmed and will be delivered on 24 June 2026.

You can view your order details by logging in to your account on [our website](#).

If you have any questions, feel free to contact our support team.

Thank you,
NorthStar Orders Team

✓ This email is from the official domain, personalized, and contains no urgent requests or suspicious links.

2. SUSPICIOUS EMAIL

From: support@northstar-security.example ⚠

To: keerit.kapoor@example.com

Subject: Important: Verify Your Account

Date: 20 June 2026, 02:15 PM

★ NORTHSTAR

Dear Customer,

We detected unusual activity on your account. Please verify your account to avoid restrictions.

Verify Your Account

<https://northstar-security.example/verify>

If you do not verify, your account may be limited.

Thank you,
NorthStar Security Team

⚠ The sender uses a lookalike domain ([northstar-security.example](mailto:support@northstar-security.example)) and creates a sense of urgency to prompt action.

3. PHISHING EMAIL

From: secure-update@payment-alert.example ✖

To: keerit.kapoor@example.com

Subject: Urgent: Payment Failed – Action Required

Date: 20 June 2026, 06:45 PM

⚠ URGENT ACTION REQUIRED

Dear Customer,

We were unable to process your payment. Your account will be suspended within 24 hours if you do not update your payment details.

Update Payment Now

<http://payment-alert.example/update>

Failure to act will result in permanent account suspension.

Thank you,
Payment Alerts Team

✖ Unrelated domain, urgent/threatening language, requests immediate action, and uses a suspicious link.

QUICK COMPARISON	Safe Email	Suspicious Email	Phishing Email
Indicator			
Sender Domain	Matches official brand domain	Lookalike domain	Unrelated / fake domain
Greeting	Personalized	Generic	Generic
Language / Tone	Professional, Informative	Cautious, Creates urgency	Urgent, Threatening
Links	Links to official website (no direct login)	Redirects to lookalike site	Suspicious link to fake site
Request for Sensitive Info	No	Implicit (verify account)	Yes (update payment details)
Overall Classification	✓ SAFE	⚠ SUSPICIOUS	✖ PHISHING

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Always think before you click. When in doubt, verify the sender, check the full email address, hover over links, and contact the organization using official channels.

✓

Be Aware. Be Secure.

Figure 1. Simulated examples demonstrating the indicators of Safe, Suspicious, and Phishing emails.

Interpretation

The safe example is informational and personalised. The suspicious example includes a lookalike domain and urgency. The phishing example combines multiple red flags: a fake domain, a payment threat, pressure to act, and a suspicious action link.

4. Phishing Indicator Analysis

The following guide converts the simulated examples into a practical checklist. It compares the indicators side by side and explains how each signal should influence the user's decision.



PHISHING INDICATOR ANALYSIS

SIMULATED TRAINING ANALYSIS — No real accounts, links, or credentials used



Indicator	SAFE EMAIL	SUSPICIOUS EMAIL	PHISHING EMAIL
 1. Sender Domain	Matches official brand domain	Lookalike / extra-word domain	Fake or unrelated domain
 2. Greeting	Personalized	Generic	Generic
 3. Tone	Professional and informative	Creates caution / mild urgency	Threatening / highly urgent
 4. Link Behavior	Official website or no action link	Requests verification via lookalike site	Pushes suspicious action link
 5. Request for Sensitive Info	No	Implicit request	Yes
 6. Pressure / Deadline	None	Limited time pressure	Immediate deadline / account threat
 7. Overall Classification	 SAFE	 SUSPICIOUS	 PHISHING

 **HOW TO ANALYZE AN EMAIL**

-  1. Check the full sender address
-  2. Read the tone and urgency
-  3. Hover over links before clicking
-  4. Look for requests for passwords, OTPs, or payment details
-  5. Verify through official channels






 Phishing attacks rely on urgency and deception. Stay alert, verify details, and protect your data.


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Date: 20 June 2026


Be Aware. Be Secure.

Figure 2. Phishing indicator analysis guide based on simulated training emails.

Key Principle

Phishing is often identified by a combination of signals. One spelling error alone may not prove fraud, but a lookalike sender, urgent payment request, generic greeting, and suspicious link together should be treated as a strong warning.

5. Risk Explanation and Response Guidance

Phishing messages commonly aim to obtain passwords, one-time codes, payment information, or access to an account. They may also encourage users to open unsafe attachments or visit deceptive websites. The risk depends on what the user is asked to do and how convincing the message appears.

Potential Outcome	How It Happens	Business / User Impact	Recommended Response
Credential theft	User enters password or OTP after following a deceptive prompt.	Account takeover and possible further fraud.	Change passwords, enable MFA, and report immediately.
Payment fraud	User provides card or banking details after an urgent payment message.	Financial loss and unauthorised transactions.	Contact the payment provider through official channels.
Malware exposure	User opens an unexpected attachment or download.	Device compromise, data loss, or unauthorised access.	Disconnect if necessary and contact IT/security support.
Data disclosure	User replies with personal, work, or customer information.	Privacy harm and possible follow-on scams.	Stop communication and report the message.

Recommended Escalation Flow

STOP Do not click, download, reply, or enter details.	VERIFY Use the official website, app, or known support channel.	REPORT Forward or report the message to the relevant IT/security contact.	SECURE If action was taken, change credentials and enable MFA.
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User-Safety Reminder

A legitimate organisation may occasionally send important notices, but it should still be possible to verify the request through a trusted, independently accessed channel. Do not use the email's own links or phone numbers to verify a suspicious message.

6. Prevention and Awareness Guidelines

The most effective phishing defence is a consistent habit of pausing, verifying, and reporting. The following practices are appropriate for students, employees, and everyday users.

Verify the sender Check the entire email address and domain, not just the display name.	Avoid unexpected attachments Do not open unfamiliar files or invoices, especially when they are unexpected.
Be cautious of urgency Threats, deadlines, and pressure are common manipulation tactics.	Protect sensitive information Never send passwords, OTPs, banking information, or card details by email.
Inspect links before clicking Hover over links and compare the destination with the legitimate organisation.	Use multi-factor authentication MFA adds protection even if a password is exposed.
Treat generic greetings carefully Messages such as “Dear Customer” deserve extra caution when paired with other red flags.	Report suspicious messages Reporting helps protect other people and allows the message to be investigated.

Quick Awareness Checklist

THINK BEFORE YOU CLICK | VERIFY BEFORE YOU SHARE | REPORT WHEN IN DOUBT

Training Outcome
The aim is not to make users afraid of every email. It is to build a calm, repeatable verification habit for messages involving urgency, payments, account access, or sensitive information.

7. Conclusion

This task demonstrated how email-based attacks can be classified by reviewing sender information, language, link behaviour, and requests for sensitive information. The simulated analysis shows that phishing messages often depend on urgency and deception rather than advanced technical complexity. The safest response to a suspicious email is to avoid interacting with it, verify through an independent official channel, and report it when appropriate. Awareness, verification, and multi-factor authentication are practical defences that reduce the likelihood of account compromise and fraud.

Skills Demonstrated

- Phishing email identification and classification
- Sender-domain and link-safety analysis
- Risk communication for non-technical users
- Cybersecurity awareness documentation
- Professional report writing and GitHub portfolio organisation

Repository Contents

File / Folder	Purpose
README.md	Project overview, classification framework, and awareness guidance.
screenshots/phishing_email_examples.png	Simulated comparison of Safe, Suspicious, and Phishing emails.
screenshots/phishing_email_analysis_guide.png	Indicator analysis guide used to explain email red flags.
Phishing_Detection_Awareness_Report_Keerit_Kapoor.pdf	Final professional report for Task 2 submission.

Disclaimer

Educational Use Only

All messages, sender addresses, links, and domains presented in this report are fictional simulations. This report does not reproduce real phishing emails, access real accounts, or direct users to operational phishing content.